"Supply-Demand & Profitability Dashboard”- (Power BI)



Project OBJECTIVE

The objective of this project is to analyze supply chain and sales performance data using Power BI to identify the key drivers behind profit and loss trends. The dashboard aims to evaluate the relationship between daily product demand and availability, quantify the impact of supply shortages on overall business performance, and provide actionable insights to help stakeholders optimize inventory planning, reduce average daily losses, and improve profitability.

key Steps for This Project

1. Data Extraction

2.Study Dataset

3.Data cleaning & Modeling

4.Data processing & DAX Queries

5. Dashboard & insights

6. Export & share project

Data source – MYSQL Database & SQL server

1. Installing MYSQL Server & MYSQL Workbench.
2. Importing the data in SQL Server.
3. Applying Left Join in SQL Server to prepare the data to be used for reporting.
4. Importing the Data to Power BI Desktop from Test Environment.

DAX Queries

1. Average Availabilty Per Day = DIVIDE([Total Availability],[Total Number of Days])

2. Average Demand Per Day = DIVIDE([Total Demand],[Total Number of Days])

3. Average Loss Per Day = DIVIDE([Total Loss],[Total Number of Days])

4. Total Availability = SUM('Demand/Availability Data'[availability])

5. Total Demand = SUM('Demand/Availability Data'[demand])

6. Total Loss = SUMX(FILTER('Demand/Availability Data','Demand/Availability Data'[LOSS/PROFIT]
<0),'Demand/Availability Data'[LOSS/PROFIT]*'Demand/Availability Data'[unit_price]) * -1

7. Total Number of Days = DISTINCTCOUNT('Demand/AvailabilityData'[Order_Date_DD_MM_YYYY].[Date])

8. Total Profit = SUMX(FILTER('Demand/Availability Data','Demand/Availability
Data'[LOSS/PROFIT]>0),'Demand/Availability Data'[LOSS/PROFIT]*'Demand/Availability Data'[unit_price])

9. Total Supply Shortage = [Total Demand]-[Total Availability]

Project Insights

- ***Loss significantly outweighs profit —*** Total Loss (8M) is substantially higher than Total Profit (301K), indicating the business is currently operating at a net deficit and requires urgent attention to loss-driving factors.
- ***Consistent daily losses —*** With an Average Daily Loss of 2.97K, losses are not isolated incidents but a recurring pattern, suggesting systemic issues in supply, pricing, or demand forecasting rather than one-off events.
- ***Demand consistently outpaces availability —*** Average Demand per Day (48.65) is nearly double the Average Availability per Day (24.70), highlighting a persistent supply-demand imbalance that likely drives missed sales opportunities and customer dissatisfaction.
- ***Supply shortage is a major contributor to loss —*** A Total Supply Shortage of 61K units reflects the cumulative effect of insufficient stock/availability, directly correlating with the high total loss figure.
- ***Availability is the bottleneck, not demand —*** Since demand remains strong and stable, the core issue lies in inventory/supply chain execution — improving availability could directly convert unmet demand into profit.
- ***Opportunity for optimization —*** Bridging even a portion of the ~24-unit average daily gap between demand and availability could meaningfully reduce the 2.97K average daily loss and improve overall profitability.

Thankyou

EVRYONE

